

**MTH-641**

**•••ZE+BA File•••**

**For Final Term  
Live Paper Files**

**.**

**Made By**

**RJ Zeeshan Zafar**

**MC190400095**

**.**

**•••Special Thanks•••**

**@Ayat @Sobia**

**@Ayesha @Sania**



A mapping  $T: X \rightarrow Y$  is said to be continuous if it is continuous at ..... point of  $X$

Answer (Please select your correct option)

☒ Every

☐ atmost one

☐ exactly one

☐ atleast one

ZE+BA Group



## MTH641 Functional Analysis

Question No : 15 of 41

The dual space of  $l^1$  is .....

Answer ( Please select your correct option )

☐  $l^p$

☐  $l^\infty$

☐  $l^2$

☐  $l^1$

Start Time: 8:06 AM

11:00

Time Left



Question Summary : ( Attempted Question )

|   |   |   |   |   |   |   |   |   |    |    |
|---|---|---|---|---|---|---|---|---|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|----|----|

[Conduct Instructions](#)

Done

Start

MathType (26 days left)



## Question No : 13 of 41

A finite dimensional normed space is ..... to its second dual

Answer ( Please select your correct option )

☐ isometric

☒ isomorphic

☐ homomorphic

☐ none of these



Question No : 6 of 41

Which one of the following axioms hold for any vectors  $u, v, w$  in vector space  $X$  and  $k, a, b$  in field  $K$  ?

Answer ( Please select your correct option )

☐  $u + (-u) = (-u) + u$ , for each  $u \in X$

☐  $u + w = w - u$

☐  $k(u + v) = ku - kv$

☒  $(ab)u = a(bu)$

Start Time: 8:06 AM



Question No : 30 of 41

An orthonormal set  $M$  is an orthogonal set in an inner product space whose elements have norm .....

Answer ( Please select your correct answer )

☒ 1

☐  $>1$

☐  $<1$

☐ None of these

Start Time: 8:06 AM

30



## MTH641 Functional Analysis

Question No : 28 of 41

Inner product is the generalization of .....

Answer ( Please select your correct option )

☐ Cross product

☒ Dot product

☐ Multiplication

☐ Distance function

Start Time: 8:06 AM

109:00

Time Left



Question Summary : 1/41



Which of the following is not true for the space  $l^p$  with  $p \neq 2$  ?

Answer ( Please select your correct option )

☐ It is a complete metric space

☐ It is a complete norm space

☒ It is a complete inner product space

☐ None of these

Start Time: 8:05 AM

110:00

Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14

[Conduct Instructions](#)

Done

Start

MathType (26 days left)

Document1 - Microsoft

VUE



Any pair of vectors in an orthogonal system is \_\_\_\_\_ to each other.

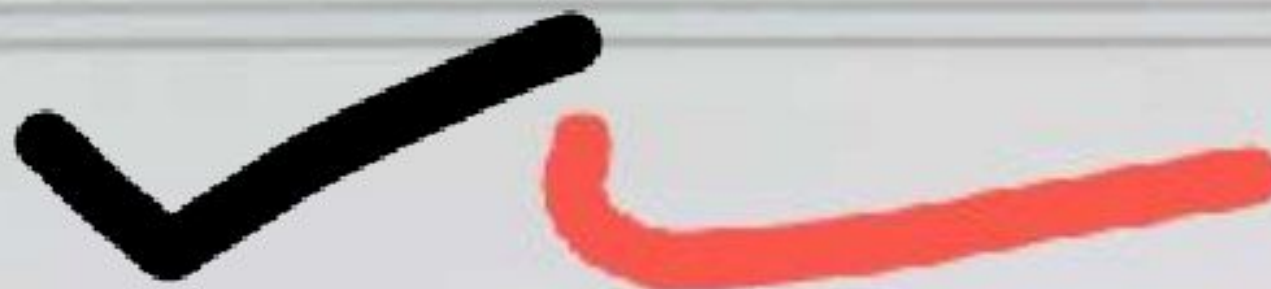
Answer ( Please select your correct option )

☐ Parallel

☐ Equal

☐ None of these

☒ Perpendicular





Question No : 10 of 41

Let  $T: D(T) \rightarrow Y$  be a linear operator, where  $D(T) \subseteq X$  and  $X, Y$  are normed spaces, then  $T$  is continuous if and only if  $T$  is \_\_\_\_\_

Answer ( Please select your correct option )

☐ continuous

☐ linear

☒ bounded

☐ none of these

Start Time: 8:06 AM

10



Two operators  $T_1$  and  $T_2$  are defined to be equal, written  $T_1 = T_2$ , if \_\_\_\_\_ and if  $T_1 x = T_2 x$  for all \_\_\_\_\_.

Answer ( Please select your correct option )

☒  $D(T_1) = D(T_2), \quad x \in D(T_1) = D(T_2)$

☐  $D(T_1) \subset D(T_2), \quad x \in D(T_1) = D(T_2)$

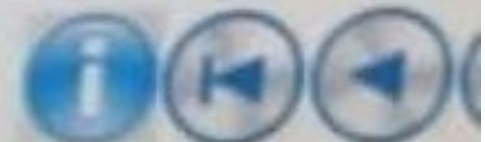
☐  $D(T_1) \supset D(T_2), \quad x \in D(T_1) = D(T_2)$

☐  $D(T_1) \supset D(T_2), \quad x \in D(T_1) \neq D(T_2)$

Start Time: 8:06 AM

112:00

12





Question No : 20 of 41

Every complete inner product space is \_\_\_\_\_ space.

Answer ( Please select your correct option )

☒ Hilbert☐ Banach☐ Unitary☐ Euclidean

Start Time 8:06 AM

111:00

Time Left



Question Summary : ( Attempted Question )

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|

[Conduct Instructions](#)

Done



Question No : 22 of 41

Which of the following is an example of an inner product space.

Answer ( Please select your correct option )

☐  $(a, b)$ ☐  $[a, b]$ ☒ Space  $\mathbb{R}^n$ ☐ None of these

Start Time: 8:06 AM

110:00

Time Left

Question Summary : ( Attempted Question  )

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

Conduct Test



Which of the following is true about the inner product space?

Answer ( Please select your correct option )

☐ Every normed space is an inner product space.

☒ Every inner product space is a normed space.

☐ Every inner product space is Hilbert space.

☐ None of these is true.

Start Time: 8:06 AM

111:00

Time Left



Question Summary : ( Attempted Question ☒ )

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|

[Conduct Instructions](#)



The dual space of a normed space  $X$  contains all ..... on  $X$

Answer ( Please select your correct option )

☒ Bounded linear functional

☐ Unbounded linear functional

☐ Bounded linear operator

☐ Unbounded linear operator

Start Time: 5:06 AM



Which of the following is a condition of an inner product space?

Answer ( Please select your correct option )

☐  $\langle x, y \rangle = \overline{\langle x, y \rangle}$

☐  $\langle \alpha x, y \rangle = \langle x, \alpha y \rangle$

☐  $\langle \alpha x + y, z \rangle = \langle x, z \rangle + \alpha \langle y, z \rangle$

☒  $\langle x, y \rangle = \overline{\langle y, x \rangle}$

Start Time: 11:00 AM

111:00

Time Left

Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Conduct Instructions



In an inner product space  $V$ , the function  $\| \cdot \| : V \rightarrow \mathbb{R}^+$  given by \_\_\_\_\_ defines a \_\_\_\_\_ in  $V$ .

Answer ( Please select your correct option )

☐  $\|x\| = \sqrt{\langle x, y \rangle}, x, y \in V$ , norm

☒  $\|x\| = \sqrt{\langle x, x \rangle}, x \in V$ , norm

☐  $\|x\| = \sqrt{\langle x, x \rangle}, x \in V$ , metric

☐ None of these

Start Time: 8:06 AM



Question No : 8 of 41

If a normed space  $X$  is \_\_\_\_\_ dimensional, then every linear operator on  $X$  is bounded.

Answer ( Please select your correct option )

☐ infinite

☒ finite

ZE+BA Group





## Question No : 26 of 41

For a subset  $A$  of an inner product space  $X$ ,  $x$  is orthogonal to  $A$  if .....

Answer ( Please select your correct option )

☐  $x$  is orthogonal to  $a$  for all  $a \in A$

☐  $x$  is orthogonal to any two elements  $a$  and  $b \in A$

☐  $x$  is orthogonal to  $a$  for a fixed  $a \in A$

☐ None of these



For any non-empty subset  $A$  of  $(X, d)$ , the diameter  $\delta(A)$  of  $A$  is defined as

Answer (Please select your correct option)

☐  $\delta(A) = \inf_{x, y \in A} d(x, y)$

☐  $\delta(A) = d(x, y)$

☒  $\delta(A) = \sup_{x, y \in A} d(x, y)$

☐  $\delta(A) = d(x, y)$

ZE+BA Group



Start Time: 11:00 AM

11:00



A sequence  $(x_n)$  in a metric space is said to be non-decreasing if for all  $n$ ,

Answer : Please select your correct option

☐  $x_n \geq x_{n+1}$

☒  $x_n \leq x_{n+1}$

☐  $x_n < x_{n+1}$

☐  $x_n > x_{n+1}$



Question No : 1 of 41

If  $d$  is a metric on  $X$ , Then for all  $x, y \in X$ ,  $d(x, y)$  is.....

Answer ( Please select your correct option )

☒ A non negative real number

☐ A negative real number

☐ Not a rational number

☐ A complex number



Question No : 16 of 41

Which of the following is a condition of an inner product space?

Answer ( Please select your correct option )

☒  $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$

☐  $\langle \alpha x + y, z \rangle = \langle x, z \rangle + \alpha \langle y, z \rangle$

☐  $\langle \alpha x, y \rangle = \langle x, \alpha y \rangle$

☐  $\langle x, y \rangle = \overline{\langle x, y \rangle}$

Start Time: 8:06 AM

111:00

Time Left

Question Summary : ( Attempted Question ☒ )

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|

Conduct Instructions



Address: <http://server/votes/test/Paper.aspx> Marks: 1 (Budgeted Time)

**MTH641 Functional Analysis**

Question No : 7 of 41

If  $S = \{v_1, v_2, v_3, \dots, v_n\}$  is a set of vectors in a finite dimensional vector space  $X$ , then  $S$  is called a basis for  $V$  if \_\_\_\_\_ and  $S$  is linearly independent.

Answer ( Please select your correct option )

- ☐ S does not span  $V$
- ☐ S is linearly dependent
- ☐ S is set of scalars
- ☒ S spans  $V$

✓

Time: 8:06 AM



If a normed space  $X$  is finite dimensional, then every linear operator on  $X$  is \_\_\_\_\_

Answer ( Please select your correct option )

☐ finite

☐ unbounded

☒ bounded

☐ none of these

Start Time: 9:05 AM

113:00

Time Left



ZE+BA Group





Every complete normed space is \_\_\_\_\_ space.

Answer ( Please select your correct option )

☒ Hilbert

☐ Banach

☐ Unstary

☐ Eucclidean

Start Time: 8:06 AM

111:00

Time Left



Question Summary : ( Attempted Question )

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|

[Conduct Instructions](#)



For subsets  $A$  and  $B$  of an inner product space  $X$ ,  $A$  is orthogonal to  $B$  if .....

Answer ( Please select your correct option )

☒  $a$  is orthogonal to  $b$  for all  $a \in A$  and  $b \in B$

☐  $a$  is orthogonal to  $b$  for a fixed  $a \in A$  and a fixed  $b \in B$

☐  $a$  is orthogonal to  $b$  for a fixed  $a \in A$  and  $b \in B$

☐  $a$  is orthogonal to  $b$  for all  $a \in A$  and a fixed  $b \in B$



Question No : 4 of 41

In a metric space  $(X, d)$  every convergent sequence is

Answer ( Please select your correct option )

☐ Unbounded sequence

☐ Not a Constant sequence

☐ Oscillating sequence

☒ A Cauchy Sequence

Start Time: 11:06 AM

114:00



Question No : 25 of 41

Isomorphism of an inner product space onto an inner product space over the same field is a ..... linear operator that preserves inner product.

Answer ( Please select your correct option )

☒ Bijective

☐ Injective

☐ Surjective

☐ None of these

Start Time: 8:05 AM

1:10:00

25



Question No : 11 of 41

Let  $T$  be a bounded linear operator. Then  $x_n \rightarrow x$  (where  $x_n, x \in D(T)$ ) implies \_\_\_\_\_

Answer (Please select your correct option)

☐  $x_n \rightarrow x$

☒  $Tx_n \rightarrow Tx$

☐  $Tx \rightarrow Tx_n$

☐ None of these

Start Time: 11:06 AM

112:00





Question No : 7 of 41

If  $S = \{v_1, v_2, v_3, \dots, v_n\}$  is a set of vectors in a finite dimensional vector space  $V$ , then  $S$  is called a basis for  $V$  if \_\_\_\_\_ and  $S$  is linearly independent.

Answer (Please select your correct option)

☐  $S$  does not span  $V$ ☐  $S$  is linearly dependent☐  $S$  is set of scalars☒  $S$  spans  $V$



## MTH641 Functional Analysis

Question No : 28 of 41

$P_n$  is the collection of all .....

Answer ( Please select your correct option )

☐ Polynomials of degree greater then equal to 'n'

☐ Polynomials of degree less then equal to 'n'

☒ Polynomials of degree equal to 'n'

☐ Polynomials of degree less then 'n'

Start Time 11:20 AM

110:00

Time Left





## MTH641 Functional Analysis

Question No : 29 of 41

The pair  $(V, \langle \cdot, \cdot \rangle)$  is called ..... space

Answer ( Please select your correct option )

☐ Norm

☐ Metric

☒ Inner product

☐ Vector

ZE+BA Group





**MTH641** Functional Analysis

Question No : 16 of 41

Which of the following is a condition of an inner product space?

Answer ( Please select your correct option )

☐  $\langle \alpha x, y \rangle = \langle x, \alpha y \rangle$

☐  $\langle \alpha x + y, z \rangle = \langle x, z \rangle + \alpha \langle y, z \rangle$

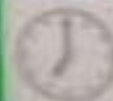
☐  $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$

☐  $\langle x, y \rangle = \overline{\langle x, y \rangle}$

Start Time: 11:20 AM

**114:00**

Time Left



Question Summary : ( Attempted Question )

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

[Conduct Instructions](#)[Click here to FINISH Your Exam](#)



Question No : 23 of 41

Which of the following is not true for the space  $C[a, b]$  with norm defined by  $\|x\| = \max_{t \in [a, b]} |x(t)|$  ?

Answer ( Please select your correct option )

☐ It is a complete metric space

☒ It is a complete norm space

☐ It is a complete inner product space

☐ It is a vector space

Start Time: 11:20 AM

111:00

Time Left

Question Summary : ( Attempted Question )

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23



A \_\_\_\_\_  $Y$  of a Banach space  $X$  is \_\_\_\_\_ if and only if the set  $Y$  is closed in  $X$ .

Answer ( Please select your correct option )

☒ subspace , complete

☐ space, complete

☐ subspace, open

☐ space, open

Start Time: 11:20 AM

115:00

Time Left



Question Summary : ( Attempted )



Every metric space is a not \_\_\_\_\_ space.

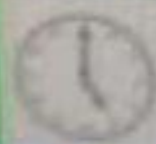
Answer ( Please select your correct option )

- ☒ a normed
- ☐ a topological
- ☐ both of these
- ☐ None of these

Start Time: 11:20 AM

113:00

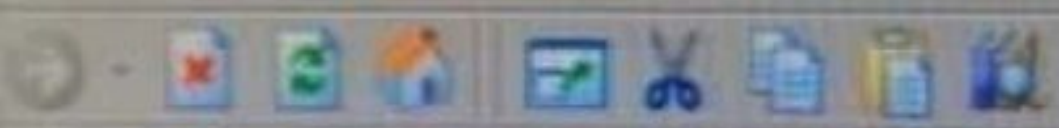
Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10





Address <http://server/vutes/client/Paper.aspx>

**MTH641** Functional Analysis

Question No : 17 of 41

Which of the following is a condition of an inner product space?

Answer ( Please select your correct option )

- ☒  $\langle x, x \rangle \geq 0$
- ☐  $\langle \alpha x + y, z \rangle = \langle x, z \rangle + \alpha \langle y, z \rangle$
- ☐  $\langle \alpha x, y \rangle = \langle x, \alpha y \rangle$
- ☐  $\langle x, y \rangle = \overline{\langle x, y \rangle}$

Start Time: 11:20 AM  
**113:00**  
Time Left

Question Summary : ( Attempted Question )

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|



Address <http://server/notes/client/Paper.aspx>

**MTH641** Functional Analysis

Question No : 24 of 41

The space ..... is not a Hilbert space

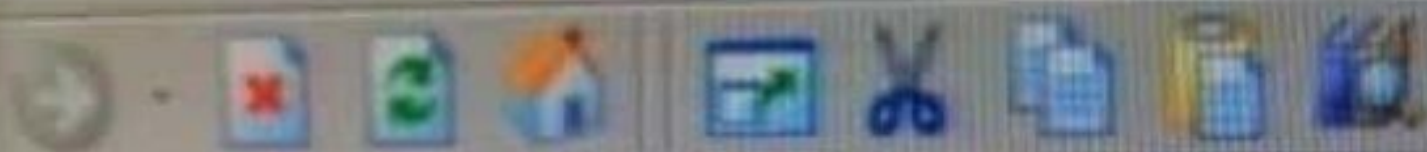
Answer ( Please select your correct option )

- ☒  $l^p$  with  $p \neq 2$
- ☐ Euclidean space  $R^n$
- ☐ Unitary Space  $C^n$
- ☐  $l^p$  with  $p = 2$

Start Time: 11:20 AM

111:00





Address <http://server/vutes/client/Paper.aspx>

## MTH641 Functional Analysis

Question No : 30 of 41

Inner product is the generalization of .....

Answer ( Please select your correct option )

☐ Cross product

☒ Dot product

☐ Multiplication

☐ Distance function

Start Time: 11:20 AM

109:00

Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7





Address <http://server/votes/ident/Paper.aspx>

## MTH641 Functional Analysis

Question No : 18 of 41

Which of the following is true about the inner product space?

Answer ( Please select your correct option )

☐ Every normed space is an inner product space.

☒ Every inner product space is a normed space.

☐ Every inner product space is Hilbert space.

☐ None of these is true.

Start Time: 11:20 AM

113:00

Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

[Conduct Instructions](#)

[Click here to](#)



File Edit View Tools Help

Address: <http://server/vutes/client/Paper.aspx>

**MTH641 Functional Analysis**

Question No : 10 of 41

A linear functional  $f$  with domain  $D(f)$  in a normed space is continuous if and only if  $f$  is \_\_\_\_\_.

Answer ( Please select your correct option )

- ☐ continuous
- ☒ bounded
- ☐ linear
- ☐ non linear

Start Time: 11:20 AM

**115:00** Time Left

Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

[Conduct Instructions](#)

[Click here to FINISH Your Exam](#)



**MTH641** Functional Analysis

Question No : 3 of 41

Given a point  $x_0 \in X$  and a real number  $r > 0$ , which of the following defines a closed ball?

Answer ( Please select your correct option )

☐  $\tilde{B}(x_0, r) = \{x \in X \mid d(x, x_0) < r\}$

☐  $\tilde{B}(x_0, r) = \{x \in X \mid d(x, x_0) > r\}$

☒  $\tilde{B}(x_0, r) = \{x \in X \mid d(x, x_0) \leq r\}$

☐  $\tilde{B}(x_0, r) = \{x \in X \mid d(x, x_0) = r\}$

Start Time: 11:20 AM

**116:00**

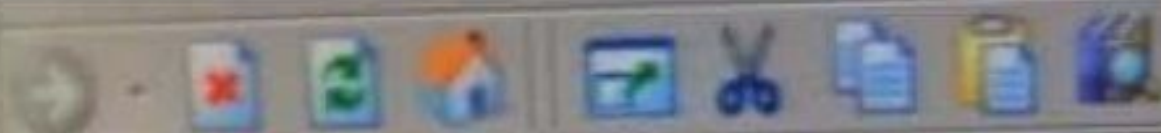
Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10





Address <http://server/vutes/client/Paper.aspx>

## MTH641 Functional Analysis

Question No : 11 of 41

The space  $C[a, b]$  is \_\_\_\_\_

Answer ( Please select your correct option )

- ☐ collection of all functions defined on  $[a, b]$
- ☐ collection of all continuous functions defined on  $(a, b)$
- ☒ collection of all continuous functions defined on  $[a, b]$
- ☐ collection of all functions defined on  $(a, b)$

Start Time: 11:20 AM

**115:00**

Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

Conduct Instructions



**MTH641** Functional Analysis

Question No : 12 of 41

Let  $X$  be an  $n$ -dimensional vector space and  $X^*$  be its dual space. Then \_\_\_\_\_

Answer ( Please select your correct option )

☐  $\dim X^* = \dim X = n$

☐  $\dim X = n$

☐  $\dim X^* = n$

☐  $\dim X^* = \dim X = m$

Start Time: 11:20 AM

**114:00**

Time Left

Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

[Contact instructions](#)

[Click here to finish your Exam](#)



## MTH641 Functional Analysis

Question No : 22 of 41

Which of the following is an example of an inner product space.

Answer ( Please select your correct option )

☐  $(a, b)$

☐  $[a, b]$

☐ Space  $R^n$

☐ None of these

Start Time: 11:20 AM

111:00



**MTH641** Functional Analysis

Question No : 2 of 41

For any three points  $x, y$  and  $z$  in a metric space  $(X, d)$ , the following property is called .....

$$d(x, z) \leq d(x, y) + d(y, z)$$

Answer ( Please select your correct option )

- ☒ Triangular inequality
- ☐ Rectangle inequality
- ☐ Holder's inequality
- ☐ Minkowski's inequality

Start Time: 11:20 AM

**116:00**

Time Left

Question Summary - ( Answered )



Address <http://server/votes/client/Paper.aspx>

**MTH641 Functional Analysis**

Question No : 7 of 41

A metric space  $X$  is said to be compact if every sequence in  $X$  has a \_\_\_\_\_ subsequence.

Answer ( Please select your correct option )

- ☐ Non decreasing
- ☐ Divergent
- ☐ Non increasing
- ☐ Convergent

Start Time: 11:20 AM

**115:00** Time Left

Question Summary : ( Attempted Question )

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

[Click here to FINISH Your Exam](#)



Address <http://server/vutes/client/Paper.aspx>

**MTH641** Functional Analysis

Question No : 9 of 41

Let  $T$  be a bounded linear operator. Then the \_\_\_\_\_ is closed.

Answer ( Please select your correct option )

☐ Banach space

☐ null space  $N(T)$

☐ Hilbert space

☐ None of these

Start Time: 11:20 AM

**115:00**  
Time Left

Question Summary



Question No : 14 of 41

IFT:  $XY$  is an Isometric Isomorphism between two norm spaces  $X$  and  $Y$ , then it implies that.....

Answer ( Please select your correct option )

☒ T is bijective and norm preserving

☐ T is injective and norm preserving

☐ T is surjective and norm preserving

☐ T is norm preserving

Start Time: 11:20 AM

114:00

Time Left



14

Question Summary : ( Attempted Question )

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

[Conduct Instructions](#)

[Click Here to Finish Your Exam](#)



Let  $\{x_1, \dots, x_n\}$  be \_\_\_\_\_ set of vectors in a normed space  $X$ . Then there is a number  $c > 0$  such that for every choice of scalars  $\alpha_1, \dots, \alpha_n$ , we have  $|\alpha_1 x_1 + \dots + \alpha_n x_n| \geq c(|\alpha_1| + \dots + |\alpha_n|)$

Answer (Please select your correct)

☐ linearly independent

☐ linearly dependent

☐ orthogonal

☐ orthonormal

Start Time: 11:20 AM

115:00

Time Left

Question Summary: ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33

Conduct Instructions

[Click here to Finish Your Exam](#)

confirm Now

topic#45

linear independent



## MTH641 Functional Analysis

Question No : 21 of 41

Every complete normed space is \_\_\_\_\_ space.

topic#82

Answer ( Please select your correct option )

☒ Hilbert

☐ Banach

☐ Unitary

☐ Euclidean

Start Time: 11:20 AM

111:00



Let  $X$  be a finite dimensional vector space. If  $x_0 \in X$  has the property that  $f(x_0) = 0$  for all  $f \in X^*$ , then \_\_\_\_\_

topic#76

$x_0 = 0$

Answer (Please select your correct answer)

☒  $x_0 = 0$

☐  $x_0 \neq 0$

☐  $x_0 = 1$

☐  $x_0 \neq 1$

ZE+BA Group

Start Time: 11:20 AM

114:00

Time Left



Question Summary: ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

[Click here to finish this exam](#)



The dual space of a normed space  $X$  contains all \_\_\_\_\_ on  $X$ .

Answer ( Please select your correct option )

☐ Bounded linear functional

☐ Unbounded linear functional

☐ Bounded linear operator

☐ Unbounded linear operator

Start Time: 11:20 AM

114:00

Time Left



15

Question Summary : ( Attempted Question )

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

Contact Instructions



Question No : 19 of 41

Every inner product space is a \_\_\_\_\_ space.

every inner product space is a normed space by .

Answer ( Please select your correct option )

☐ Metric

☐ Normed

☐ Topological

☒ All of above

topic#88

Start Time: 11:20 AM

11:20:00



Question No : 1 of 41

The property  $d(x, y) \geq 0$  of a metric space is called .....

Answer ( Please select your correct option )

☒ Non negativity

☐ Reflexive property

☐ Symmetric property

☐ Triangle property

Start Time: 11:20 AM

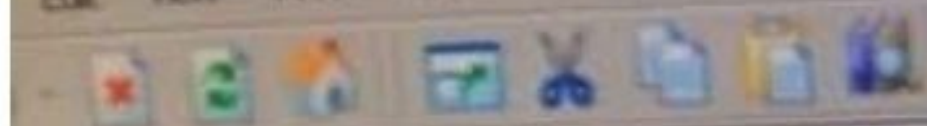
116:00

Time Left



Question Summary : ( Attempted Question )





## ATH641 Functional Analysis

Marks: 1 (Bu

Question No : 25 of 41

Isomorphism of an inner product space onto an inner product space over the same field is a ..... linear operator that preserves inner product.

Answer (Please select your correct option)

☐ Bijective☐ Injective☐ Surjective☐ None of these

Start Time: 11:20 AM

110:00

Time Left

Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10





Address <http://server/vutes/client/Paper.aspx>

## MTH641 Functional Analysis

Question No : 4 of 41

Two metric spaces  $(X, d)$  and  $(\tilde{X}, \tilde{d})$  are said to be isometric if there exists ..... isometry of  $X$  onto  $\tilde{X}$

Answer ( Please select your correct option )

☐ An injective

☒ A bijective

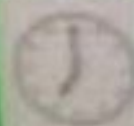
☐ A surjective

☐ none of these

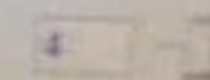
Start Time 11:20 AM

115:00

Time Left



Question Summary : ( Attempted Question )





Let  $T: D(T) \rightarrow Y$  be a linear operator, where  $D(T) \subseteq X$  and  $X, Y$  are normed spaces, then  $T$  is \_\_\_\_\_ if and only if  $T$  is bounded.

Answer (Please select your correct

☒ continuous

☐ linear

☐ bounded

☐ none of these

Start Time: 11:20 AM

115:00

Time Left

Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

Conduct Instructions

[Click here to FINISH Your Exam](#)



## MTH641 Functional Analysis

Question No : 26 of 41

$\|x + y\| \leq \|x\| + \|y\|$  is called .....

Answer ( Please select your correct option )

☐ Triangle inequality

☐ Minkowski's inequality

☐ Holder's inequality

☒ Schwarz inequality

Start Time: 11:20 AM

**110:00**  
Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

[Conduct instructions](#)



For orthogonal elements  $x, y$  we have  $\langle x, y \rangle = 0$  and the Pythagorean relation is .....

Answer ( Please select your correct option )

☐  $\|x+y\|^2 = 2\|x\|^2 + 2\|y\|^2$

☐  $\|x+y\|^2 = \|x\|^2 - \|y\|^2$

☐  $\|x-y\|^2 = \|x\|^2 + \|y\|^2$

☒  $\|x+y\|^2 = \|x\|^2 + \|y\|^2$

Start Time: 11:20 AM

110:00

Time Left

Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

Conduct Instructions



A mapping  $T: X \rightarrow Y$  is said to be continuous if it is continuous at ..... point of  $X$

Answer ( Please select your correct option )

☒ Every

☐ atmost one

☐ exactly one

☐ atleast one

11:53 AM ✓



Question No : 13 of 41

A finite dimensional normed space is ..... to its second dual.

topic#76 +topic#77

Answer ( Please select your correct option )

☐ isometric

☒ isomorphic

☐ homomorphic

☐ none of these

11:53 AM ✓



Question No : 30 of 41

An orthonormal set  $M$  is an orthogonal set in an inner product space whose elements have norm .....

Answer ( Please select your correct option )

☒ equal to 1

☐  $>1$

☐  $<1$

☐ None of these

11:53 AM ✓✓

Start Time: 8:06 AM

30



Question No : 22 of 41

Which of the following is an example of an inner product space.

Answer ( Please select your correct option )

☐  $(a, b)$

☐  $[a, b]$

☒ Space  $R^n$

☐ None of these

Start Time: 8:06 AM

110:00

Time Left



Question Summary : ( Attempted Question )

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

11:53 AM ✓



In an inner product space  $V$ , the function  $\| \cdot \| : V \rightarrow \mathbb{R}^+$  given by \_\_\_\_\_ defines a \_\_\_\_\_ in  $V$

Answer ( Please select your correct option )

☐  $\|x\| = \sqrt{\langle x, y \rangle}, x, y \in V$ , norm

☐  $\|x\| = \sqrt{\langle x, x \rangle}, x \in V$ , norm

☐  $\|x\| = \sqrt{\langle x, x \rangle}, x \in V$ , metric

☐ None of these

11:53 AM ✓✓

Start Time 8:06 AM



A sequence  $(x_n)$  in a metric space is said to be non-decreasing if for all  $n$ ,

Answer ( Please select your correct option )

☐  $x_n \geq x_{n+1}$

☒  $x_n \leq x_{n+1}$

☐  $x_n < x_{n+1}$

☐  $x_n > x_{n+1}$

11:53 AM ✓



Question No : 1 of 41

If  $d$  is a metric on  $X$ , Then for all  $x, y \in X$ ,  $d(x, y)$  is....

Answer ( Please select your correct option )

☒ A non negative real number

☐ A negative real number

☐ Not a rational number

☐ A complex number

11:53 AM ✓



MTH641 Functional Analysis

Question No : 7 of 41

If  $S = \{v_1, v_2, v_3, \dots, v_n\}$  is a set of vectors in a finite dimensional vector space  $X$ , then  $S$  is called a basis for  $V$  if \_\_\_\_\_ and  $S$  is linearly independent.

Answer (Please select your correct option)

☐  $S$  does not span  $V$

☐  $S$  is linearly dependent

☐  $S$  is set of scalars

☒  $S$  spans  $V$

ZE+BA Group



11:53 AM



Start Time: 9:06 AM



Question No : 13 of 41

A finite dimensional normed space is ..... to its second dual

Answer ( Please select your correct option )

☐ isometric

☒ isomorphic

☐ homomorphic

☐ none of these

12:11 PM ✓



Every metric space is a not \_\_\_\_\_ space.

Answer ( Please select your correct option )

☐ a normed

☐ a topological

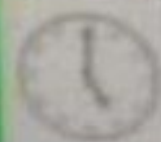
☐ both of these

☐ None of these

Start Time: 11:20 AM

113:00

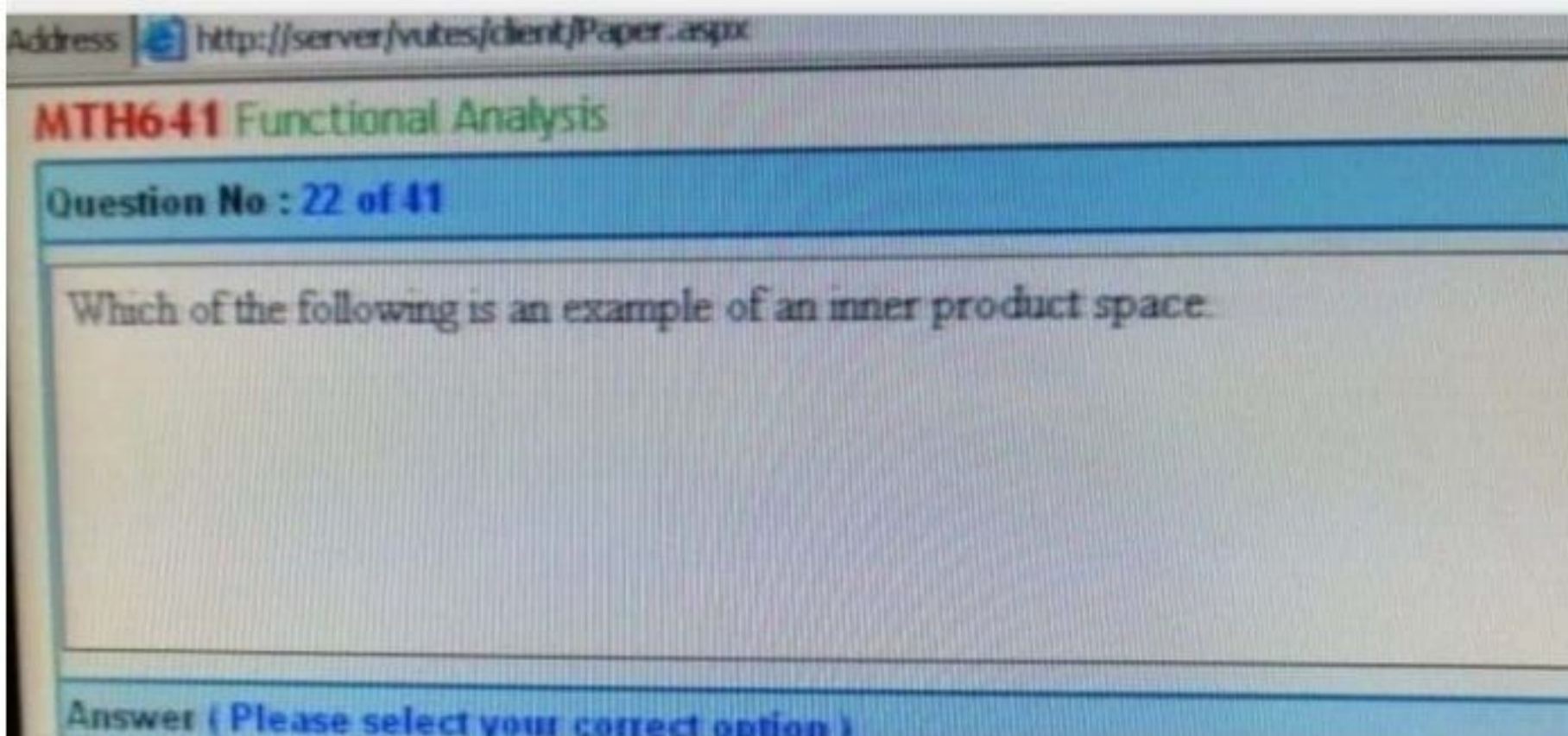
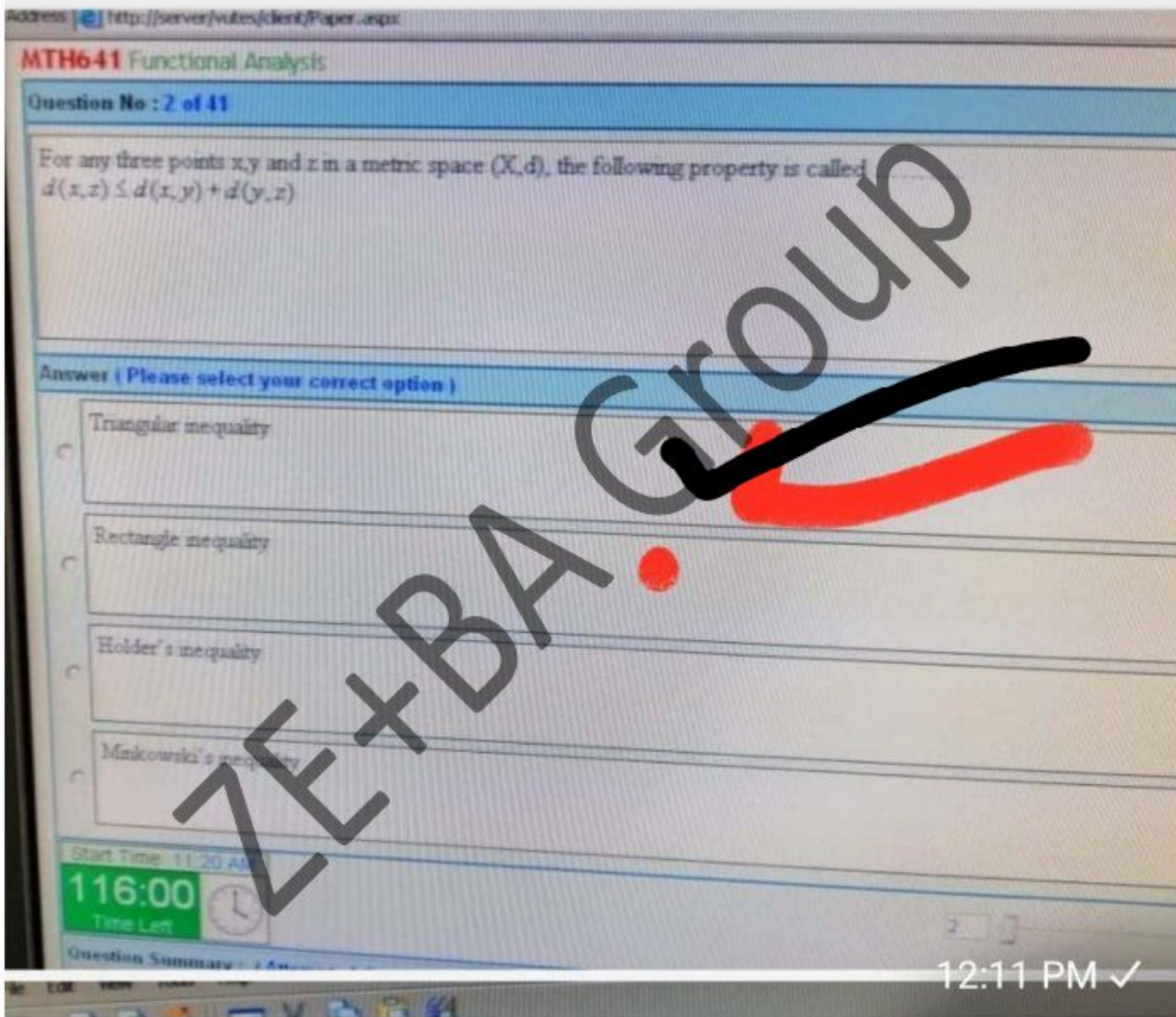
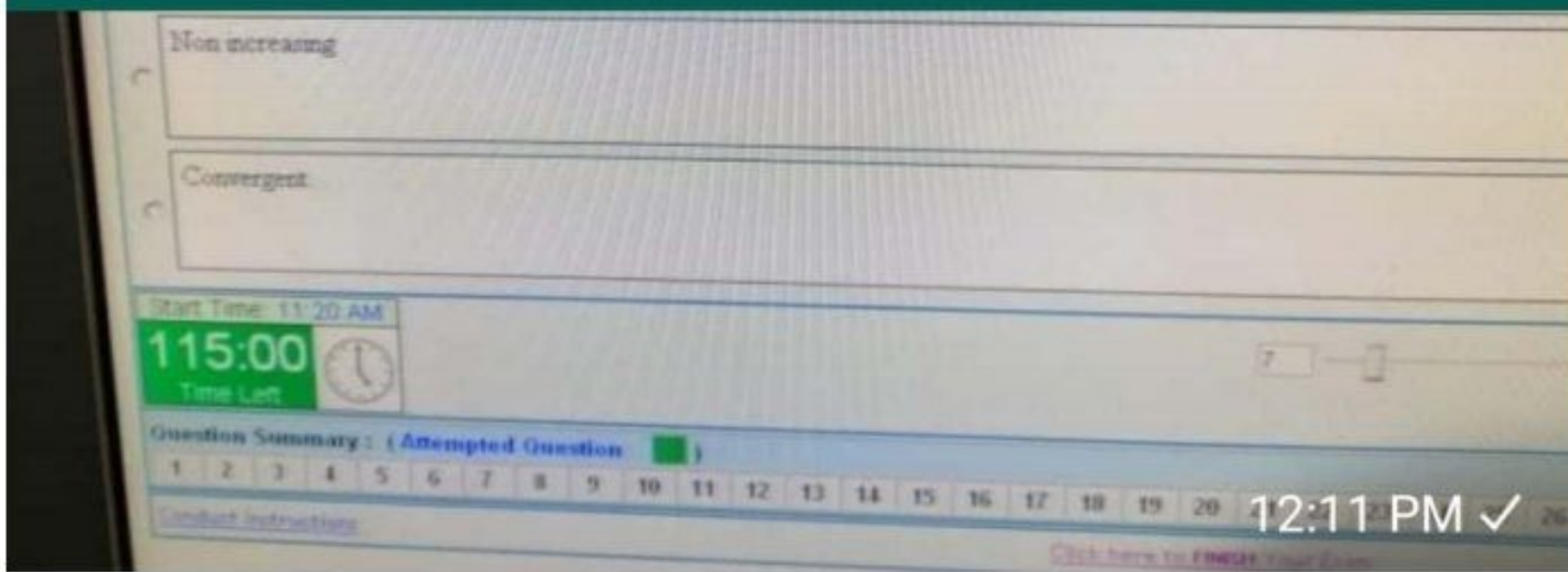
Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11







Address <http://server/vutes/client/Paper.aspx>

**MTH641 Functional Analysis**

Question No : 7 of 41

A metric space  $X$  is said to be compact if every sequence in  $X$  has a \_\_\_\_\_ subsequence.

Answer ( Please select your correct option )

- ☐ Non decreasing
- ☐ Divergent
- ☐ Non increasing
- ☐ Convergent

Start Time: 11:20 AM

**115:00**  
Time Left

Question Summary : ( Attempted Question )

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

Conduct Instructions

12:11 PM ✓



Address: http://server/vutes/client/vaper.aspx

## MTH641 Functional Analysis

Question No : 6 of 41

Marks: 1 (Budge

Let  $\{x_1, \dots, x_n\}$  be \_\_\_\_\_ set of vectors in a normed space  $X$ . Then there is a number  $c > 0$  such that for every choice of scalars  $\alpha_1, \dots, \alpha_n$ , we have

$$\|\alpha_1 x_1 + \dots + \alpha_n x_n\| \geq c(|\alpha_1| + \dots + |\alpha_n|)$$

Answer ( Please select your correct option )

☐ linearly independent☐ linearly dependent☐ orthogonal☐ orthonormal

Start Time: 11:20 AM

115:00

Time Left

Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33

[Conduct Instructions](#)[Click here to FINISH Your Exam](#)

12:11 PM ✓



Let  $X$  be a finite dimensional vector space. If  $x_0 \in X$  has the property that  $f(x_0) = 0$  for all  $f \in X^*$ , then \_\_\_\_\_

Answer ( Please select your correct option )

☐  $x_0 = 0$

☐  $x_0 \neq 0$

☐  $x_0 = 1$

☐  $x_0 \neq 1$

Start Time: 11:20 AM

114:00

Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

Contact instructions

[Click here to finish your Exam](#)

12:11 PM ✓



Question No : 19 of 41

Every inner product space is a \_\_\_\_\_ space.

Answer ( Please select your correct option )

☐ Metric

☐ Normed

☐ Topological

☒ All of above

Start Time: 11:20 AM

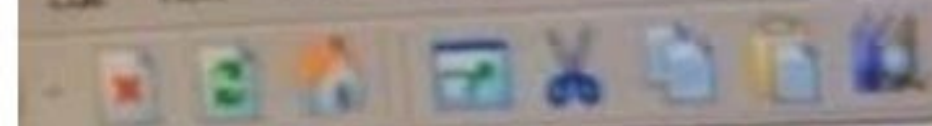
11:20:00

12:11 PM ✓

Scanned with CamScanner

Scanned with CamScanner





Isomorphism of an inner product space onto an inner product space over the same field is a ..... linear operator that preserves inner product

Answer ( Please select your correct option )

☒ Bijective

☐ Injective

☐ Surjective

☐ None of these

Start Time: 11:00 AM

110:00

Time Left

Question Summary : ( Attempted 2 Question )

1 2 3 4 5 6 7 8 9 10

12:11 PM ✓



Address <http://server/vutes/client/Paper.aspx>

## MTH641 Functional Analysis

Question No : 26 of 41

$\|x+y\| \leq \|x\| + \|y\|$  is called .....

Answer ( Please select your correct option )

☐ Triangle inequality

☐ Minkowski's inequality

☐ Holder's inequality

☐ Schwarz inequality

Start Time 11:20 AM

**110:00**

Time Left



Question Summary : ( Attempted Question )

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

[Conduct Instructions](#)

12:11 PM ✓



Question No : 7 of 11

If  $S = \{v_1, v_2, v_3, \dots, v_k\}$  is a set of vectors in a finite dimensional vector space  $X$ , then  $S$  is called a basis for  $V$  if \_\_\_\_\_ and  $S$  is linearly independent.

Answer : (Please select your correct option)

☐  $S$  does not span  $V$ ☐  $S$  is linearly dependent☐  $S$  is set of scalars☒  $S$  spans  $V$ 

12:11 PM ✓



Question No : 11 of 41

Let  $T$  be a bounded linear operator. Then  $x_n \rightarrow x$  where  $x_n, x \in D(T)$  implies \_\_\_\_\_

Answer ( Please select your correct option )

☐  $x_n \rightarrow x$

☒  $Tx_n \rightarrow Tx$

☐  $Tx \rightarrow Tx_n$

☐ None of these

Start Time: 8:06 AM

112:00



12:11 PM ✓



Question No : 27 of 41

For subsets  $A$  and  $B$  of an inner product space  $X$ ,  $A$  is orthogonal to  $B$  if .....

Answer ( Please select your correct option )

☒  $a$  is orthogonal to  $b$  for all  $a \in A$  and  $b \in B$

☐  $a$  is orthogonal to  $b$  for a fixed  $a \in A$  and a fixed  $b \in B$

☐  $a$  is orthogonal to  $b$  for a fixed  $a \in A$  and  $b \in B$

☐  $a$  is orthogonal to  $b$  for all  $a \in A$  and a fixed  $b \in B$

12:11 PM ✓



Question No : 9 of 41

If a normed space  $X$  is finite dimensional, then every linear operator on  $X$  is \_\_\_\_\_

Answer ( Please select your correct option )

☐ finite

☐ unbounded

☒ bounded

☐ none of these

Start Time: 8:06 AM

113:00

Time Left



12:11 PM ✓



Two operators  $T_1$  and  $T_2$  are defined to be equal, written  $T_1 = T_2$ , if \_\_\_\_\_ and if  $T_1 x = T_2 x$  for all \_\_\_\_\_

Answer ( Please select your correct option )

☒  $D(T_1) = D(T_2), \quad x \in D(T_1) = D(T_2)$

☐  $D(T_1) \subset D(T_2), \quad x \in D(T_1) = D(T_2)$

☐  $D(T_1) \supset D(T_2), \quad x \in D(T_1) = D(T_2)$

☐  $D(T_1) \supset D(T_2), \quad x \in D(T_1) \neq D(T_2)$

Start Time: 8:05 AM

112:00

12

12:11 PM ✓



Any pair of vectors in an orthogonal system is \_\_\_\_\_ to each other.

Answer ( Please select your correct option )

Parallel

Equal

None of these

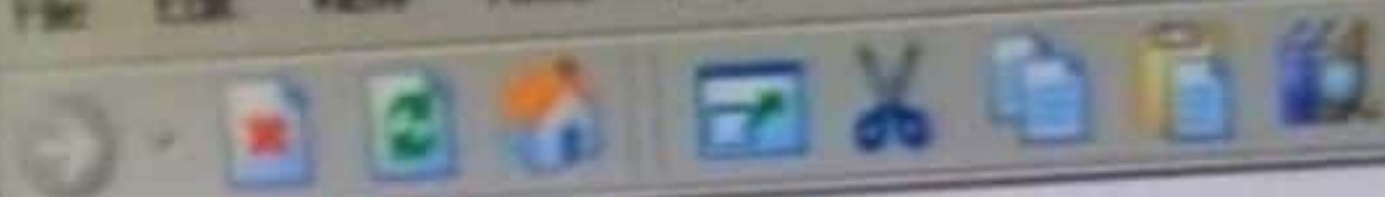
Perpendicular

ZE+BA Group



12:11 PM ✓



Address <http://server/votes/client/Paper.aspx>**MTH641** Functional Analysis

Question No : 18 of 41

Which of the following is true about the inner product space?

topic#88

Answer ( Please select your correct option )

☐ Every normed space is an inner product space.☒ Every inner product space is a normed space.☐ Every inner product space is Hilbert space.☐ None of these is true.

Start Time: 11:20 AM

**113:00**

Time Left



Question Summary : ( Attempted Question )

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

[Conduct Instructions](#)[Click here](#)

12:11 PM ✓



## MTH641 Functional Analysis

Question No : 3 of 41

Given a point  $x_0 \in X$  and a real number  $r > 0$ , which of the following defines a closed ball?

Answer ( Please select your correct option )

☐  $\tilde{B}(x_0, r) = \{x \in X \mid d(x, x_0) < r\}$

☐  $\tilde{B}(x_0, r) = \{x \in X \mid d(x, x_0) > r\}$

☒  $\tilde{B}(x_0, r) = \{x \in X \mid d(x, x_0) \leq r\}$

☐  $\tilde{B}(x_0, r) = \{x \in X \mid d(x, x_0) = r\}$

Start Time: 11:20 AM

116:00

Time Left

Question Summary : ( Attempted Question )

1

2

3

4

5

6

7

8

9

10

11

12

12:11 PM ✓